

Parallel Consensus Routing: Training-Free Quality Control, Coordinated Hallucination, and Model-Pair Selection in Cost-Efficient LLM Inference

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Abstract—Cost-efficient large language model (LLM) inference is dominated by two paradigms: *pre-query routing*, which commits to a model before any output exists, and *sequential cascading*, which escalates cheap-to-expensive in series and accrues additive latency. We study Parallel Consensus Routing (PCR), a third paradigm that runs two inexpensive models simultaneously and uses their binary output agreement as a training-free quality oracle: agreed queries are resolved immediately, disagreements trigger exactly one frontier escalation. We run a balanced four-pair design — two within-family and two cross-family cheap pairs, with the frontier model and the serving provider held constant — on a complete six-benchmark grid (ARC, OpenBookQA, CommonsenseQA, MMLU, TruthfulQA, MMLU-Pro), all 24 cells populated, scoring eight routing policies on identical stored data. Every pair reported completed all six benchmarks; pairs that finished only a subset were excluded rather than shown with gaps. Five results emerge. (1) Coordinated hallucination (C-Hall)—two cheap models agreeing on a wrong answer—scales monotonically with question difficulty, not task type: the mean rises from $\approx 3\%$ on ARC/OpenBookQA to $\approx 19\%$ on MMLU-Pro (25.9% for the within-family Qwen pair), reaching on a plain multiple-choice task the error rates previously reported only for mathematical reasoning. (2) A capability-matched cheap pair with PCR matches always-frontier accuracy on easy/medium questions at 40–70% lower cost, but on hard questions its accuracy is safe while its C-Hall (12–26%) is not. (3) Shared model lineage buys a large agreement premium on hard/adversarial benchmarks (+9 to +22 pp; CommonsenseQA, TruthfulQA, MMLU-Pro) and, suggestively, a coordinated-hallucination premium there too (TruthfulQA +6.6 pp, CI [+0.3, +12.8]); on easy/medium benchmarks neither effect appears. With two pairs per condition the C-Hall signal is a hypothesis for powered follow-up. What makes a pair usable at all is capability matching, not lineage. (4) A third cheap model (PCR-3, majority vote) reliably breaks ties but yields no accuracy or safety gain. (5) On free-tier APIs, two of {gpt-oss-20B, Qwen3.6-27B, Qwen3.8-27B} on a low-latency provider is the recommended default: solo accuracies within ~ 1 pp, ≈ 0.5 s latency, ≈ 5 –12% escalation. All experiments used free-tier inference and required \$0 marginal API spend.

Index Terms—LLM routing, cost-efficient inference, parallel consensus, coordinated hallucination, model-pair selection, capability matching, black-box API inference

I. INTRODUCTION

The LLM API marketplace is sharply tiered: frontier models command costs one to two orders of magnitude above small open models [1]. For throughput workloads, routing *all* queries to a frontier is prohibitively expensive and routing all to a cheap model forfeits quality on the hard tail. The engineering

problem is: *given a query, which model or combination minimizes cost subject to a quality constraint?*

Three paradigms exist. **Pre-query routing** [2], [3] predicts the right tier from input features before any output is generated. **Sequential cascading** [4] queries cheap first and applies a learned scorer to decide escalation, adding one model’s latency per hop. **Self-verification routing** [7] lets a small model grade its own answer, but self-verification is noisy on reasoning [13].

We study a structurally different idea: **use two cheap models as quality oracles for each other**. Two independently deployed cheap models queried in parallel that converge on the same answer provide agreement evidence that is stronger than either model’s self-assessment. On disagreement, exactly one frontier call is made. The system is training-free, black-box compatible, and faster than a serial cascade on every agreed query. The analogy to speculative decoding [5], [6] is exact—draft-then-verify at the response level rather than the token level—but without the white-box probability access speculative decoding requires.

The failure mode that makes this non-trivial is **coordinated hallucination (C-Hall)**: two cheap models agreeing on the *same wrong* answer. C-Hall is invisible to the router—it is indistinguishable from correct consensus—so understanding when it occurs is a safety prerequisite, not an optimization.

a) Contributions: This paper (i) reproduces and extends an initial single-pair PCR study with a **balanced 4-pair design on a complete 6-benchmark grid** (frontier and provider held constant), every cell populated; (ii) shows C-Hall **scales with difficulty, not task type**, generalizing the earlier “knowledge vs. math” dichotomy to a difficulty-tier one; (iii) runs a **full 8-policy ablation** (Always-Cheap, Best-Cheap, Consensus, PCR-2, Oracle-2-cheap, Random, Always-Frontier, PCR-3) on identical data; (iv) characterizes the **model-family effect**—a large agreement premium and a suggestive coordinated-hallucination premium, both on hard/adversarial benchmarks only—and identifies **capability matching** as the constraint that makes a pair usable at all; (v) provides measured cost/latency and a concrete **practitioner pair-selection guide**.

II. RELATED WORK

A. Sequential Cascade Routing

FrugalGPT [4] defines an LLM cascade with a learned (DistilBERT) accept/escalate scorer, matching GPT-4 at up to 98% cost reduction on classification, but requiring task-specific supervised training and incurring latency proportional to cascade depth. Hybrid LLM [3] predicts the small-vs-large quality gap pre-query (up to 40% fewer large calls, <1% quality drop) but cannot use observed output. AutoMix [7] is architecturally closest to PCR: a single model self-verifies before escalation, with a POMDP meta-verifier absorbing self-verification noise. Our cross-model agreement is stronger than self-verification in principle—independent models cannot share a self-delusion—though, as our matrix shows, shared training corpora still induce correlated errors.

B. Pre-Query Routing

RouteLLM [2] trains four router variants on Chatbot Arena preference data (>85% cost reduction on MT-Bench, 45% on MMLU). GraphRouter [25] models query-LLM relationships with GNNs; FORC [24] learns per-LLM score prediction via contrastive learning; RouterBench [21] standardizes evaluation. All require trained components; PCR requires none.

C. Multi-Model Consensus and Ensembles

Multi-agent debate [8] improves reasoning through iterative text exchange at $O(\text{rounds} \times \text{models})$ cost. Self-Consistency [10] majority-votes multiple samples from one model; LLM-Blender [9] fuses open-model outputs with a trained ranker; ChatEval [26] debates for evaluation. None exploit cost differentials between tiers, and none use a single-round binary agreement signal.

D. Speculative Decoding

Leviathan et al. [5] and Chen et al. [6] accelerate decoding with a draft model whose tokens the target verifies by rejection sampling (2–3 $\times$ speedup, provable equivalence). This needs the target’s per-token distribution—unavailable for proprietary APIs. PCR realizes the same draft-then-verify separation at the *response* level on final text only.

E. Semantic Caching, Calibration, and Hallucination

vCache [11] shows global similarity thresholds carry no correctness guarantee and proposes per-embedding online learning; our per-domain/per-difficulty threshold table is the routing analogue. SelfCheckGPT [12] detects hallucination via multi-sample consistency; Kadavath et al. [13] show LLMs are poorly calibrated on multi-step reasoning. Coordinated hallucination—agreement *negatively* correlated with correctness on hard items—has not been characterized in the routing literature.

F. Benchmarks

Our grid spans MMLU [14] and ARC-Challenge [16], plus TruthfulQA [17] (adversarial misconceptions), MMLU-Pro [18] (10-option, reasoning-heavy), OpenBookQA [19] and CommonsenseQA [20] — six multiple-choice sets that span a wide difficulty range while holding the answer format fixed. We exclude numeric-answer benchmarks such as GSM8K [15] from the grid: there, answer extraction interacts with the generation budget, which confounds the agreement measurement (§VI). Chain-of-thought [23] is suppressed to isolate the agreement signal.

G. Research Gap

No prior work evaluates a *parallel binary-consensus* quality oracle for cost reduction, characterizes coordinated hallucination and its difficulty-dependence, controls for model-family relatedness while holding capability constant, or provides an empirical pair-selection guide for practitioners. This paper does all four.

III. METHODOLOGY

A. Problem Formulation

Let M_A, M_B be cheap models and M_F a frontier model, with $\text{cost}(M_F) \gg \text{cost}(M_A) \approx \text{cost}(M_B)$. For a multiple-choice query q each model emits a letter $r \in \{A, \dots\}$. The **PCR-2** decision rule is

$$\text{output}(q) = \begin{cases} r_A(q) & r_A(q) = r_B(q) \quad [\text{agreement}] \\ M_F(q) & r_A(q) \neq r_B(q) \quad [\text{disagreement}] \end{cases} \quad (1)$$

with expected per-query cost $\mathbb{E}[\text{cost}] = \text{cost}(M_A) + \text{cost}(M_B) + \text{cost}(M_F) \cdot P(\text{disagree})$.

B. PCR-3: Majority Vote

PCR-3 runs three cheap models M_A, M_B, M_C ; if any two agree, their answer is taken; only if all three differ is M_F called. This trades one extra cheap call for a lower escalation rate and a majority-vote margin.

C. Policy Space

Every policy below is scored on the *same* stored per-question data (cheap answers, agreement, frontier answer on disagreements, per-call token and latency instrumentation), so no policy requires additional API calls except Always-Frontier (see §IV): *Always-Cheap-A/B* (one small model); *Best-Cheap* (the stronger of the two); *Consensus, no escalation* (trust agreement, fall back to M_A on disagreement—provably equal to Always-Cheap-A); *PCR-2*; *Oracle-2-cheap* (correct whenever *either* cheap model is correct—the ceiling for any two-cheap method); *Random* (coin flip M_A vs. M_F); *Always-Frontier*; *PCR-3*.

D. Agreement Signal and Outcome Categories

We optionally compute embedding cosine similarity $s \in [-1, 1]$ between raw responses (all-MiniLM-L6-v2 [22]). Three outcome categories: **correct consensus** ($r_A = r_B = \text{gold}$); **coordinated hallucination** ($r_A = r_B \neq \text{gold}$); **disagreement** ($r_A \neq r_B$).

TABLE I
LITERATURE REVIEW — RELATED WORK AND DIFFERENTIATION OF CONTRIBUTION

Reference	Year/Venue	Approach	Key Limitation	PCR Advantage
FrugalGPT [4]	2023 / TMLR	Sequential cascade, trained quality scorer	Labeled training data; serial latency	Training-free; parallel; bounded at one frontier call
Hybrid LLM [3]	2024 / ICLR	Pre-query difficulty-gap router	No output signal; pre-commit	Post-query observed agreement
RouteLLM [2]	2024 / arXiv	Preference-trained router (4 variants)	Human preference labels	Zero labeled data; any API pair
AutoMix [7]	2024 / ACL	Self-verify + POMDP router	Self-verification noisy; POMDP training	Cross-model verification; training-free
Multi-agent Debate [8]	2024 / ICML	Iterative multi-round debate	$O(\text{rounds} \times \text{models})$; high latency	Single-round; throughput-optimized
Self-Consistency [10]	2023 / ICLR	Majority vote, one model	Same tier; no cost cut	Cross-tier; cheap replaces expensive samples
Spec. Decoding [5], [6]	2023 / ICML	Draft-then-verify on token probs	White-box only	Black-box; final text only
vCache [11]	2025 / arXiv	Online per-embedding threshold	Caching, not routing	Per-domain/difficulty calibration for routing
SelfCheckGPT [12]	2023 / EMNLP	Multi-sample consistency	Consistency assumed positive	C-Hall: consistency as <i>negative</i> indicator on hard items
RouterBench [21]	2024 / arXiv	Routing benchmark	No algorithm proposed	Results compatible with its methodology

E. Coordinated Hallucination Hypothesis

We hypothesize C-Hall arises from shared inductive biases in overlapping training corpora: on retrieval both models recall the same fact (beneficial), on procedural or adversarial items both may pattern-match the same wrong shortcut or misconception. We test two moderators: (a) benchmark **difficulty**, and (b) whether the pair is **within-family** (same developer/lineage) or **cross-family**.

F. Capability Matching

A within-vs-cross comparison is only interpretable if the two cheap models are capability-matched: if one model is much weaker, high agreement reflects the weak model deferring to a shared plausible-wrong answer (a skill gap), not family structure. We require the two cheap models’ *solo* accuracy to differ by ≤ 3 pp on each benchmark; cells violating this are reported but excluded from family pooling.

G. Difficulty Tiers

We group the six benchmarks by observed difficulty: **easy** (ARC-Challenge, OpenBookQA, CommonsenseQA), **medium** (MMLU), and **hard** (TruthfulQA, MMLU-Pro).

IV. EXPERIMENTAL SETUP

A. Inclusion criterion

Every result in this paper comes from a pair that completed **all six benchmarks at full N** . We screened eleven candidate pairs; seven were dropped before analysis — two used models the providers delisted mid-study, two ran on only one or two benchmarks, and three were blocked partway by a provider’s free-tier cap (two of those also failed the capability-matching bar of §V-F). Reporting a pair on a subset of benchmarks would make cross-pair comparison depend on which cells

happened to finish, so those pairs are excluded entirely rather than shown with gaps. What remains is a **balanced** 4×6 **design**: two within-family pairs and two cross-family pairs, all sharing one frontier model and one provider (Table III), with 24 of 24 cells populated.

B. Models

All models are free tier. Cheap models: `gpt-oss-20b`, `gpt-oss-safeguard-20b` (both OpenAI-oss), `qwen3.6-27b`, `qwen3.8-27b` (both Alibaba Qwen3) — all ≤ 27 B and served on Groq. Frontier: `gpt-oss-120b`, identical for all four pairs, so frontier capability is held constant and only the cheap-pair lineage varies.

C. Benchmarks

Six multiple-choice benchmarks (Table II), spanning elementary science to adversarial misconceptions and 10-option reasoning. Letter extraction is generalized to A–J. TruthfulQA MC1 is reduced to 6 options (the true option plus ≤ 5 sampled distractors, shuffled per question) for comparability. Cheap answer budget is 256 tokens throughout (measured: reasoning-tuned cheap models need ~ 200 hidden tokens before emitting a letter). We deliberately restrict to a single answer format: a numeric-answer benchmark was piloted but is excluded, because answer extraction there is confounded with the token budget (§VI).

D. Model pairs

The four pairs meeting the inclusion criterion (Table III) form a 2×2 design: lineage (within-family vs. cross-family) crossed with two distinct model choices per condition. All four share the frontier (`gpt-oss-120b`) and the provider, so neither confounds the comparison. Pairs 10–11 were constructed after the solo sweep (below) identified `gpt-oss-20b`,

TABLE II
BENCHMARKS. EVERY PAIR IN THIS STUDY IS EVALUATED ON ALL SIX.

Benchmark	Options	N	Character
OpenBookQA	A–D	200	elementary science
ARC-Challenge	A–E	150–300	science reasoning
CommonsenseQA	A–E	200	commonsense
MMLU	A–D	250	general knowledge, 57 domains
TruthfulQA (MC1)	A–F	150	adversarial misconceptions
MMLU-Pro	A–J	150	hard, reasoning-heavy

TABLE III
THE FOUR PAIRS. FAMILY = WHETHER THE TWO CHEAP MODELS SHARE A DEVELOPER/LINEAGE. FRONTIER AND PROVIDER ARE CONSTANT ACROSS ALL FOUR.

Pair	Family	Cheap A	Cheap B
Pair 3	within	gpt-oss-20b	gpt-oss-safeguard-20b
Pair 7	within	qwen3.8-27b	qwen3.6-27b
Pair 10	cross	gpt-oss-20b	qwen3.6-27b
Pair 11	cross	gpt-oss-20b	qwen3.8-27b
PCR-3 trio	—	gpt-oss-20b + qwen3.6-27b + qwen3.8-27b	

Frontier for every row: gpt-oss-120b. Provider for every row: Groq.

qwen3.6-27b and qwen3.8-27b as mutually capability-matched to ~ 1 pp. A three-model trio drawn from the same set is used for the PCR-3 ablation.

E. Solo model sweep and capability gaps

Each model was run *alone* ($N = 80$ /benchmark) to measure standalone accuracy (Table IV, upper block). This provides the Always-Frontier baseline used throughout the ablation, and lets each pair’s capability gap be read directly (lower block).

Three of the four benchmarks put every pair inside the ± 3 pp matching bar of §V-F. **TruthfulQA is the exception:** gpt-oss-20b is markedly weaker there (71.2) than either Qwen model (81.2/83.8), so the two *cross*-family pairs carry 10.0 and 12.6 pp gaps on that benchmark while the within-family pairs stay at 2.6–3.8 pp. We report this rather than drop the cells, because the confound runs *against* our main family result: a mismatched pair should show *inflated* C-Hall (§V-F), yet on TruthfulQA the mismatched cross-family pairs show the *lower* C-Hall (9.5%, 11.2%) and the matched within-family pairs the higher (14.3%, 19.8%). The within-family premium of Finding 5 therefore survives a bias pointed the opposite way; it is not an artifact of mismatch.

F. Infrastructure

Python 3.13, `asyncio` for parallel cheap calls, temperature 0, seed 42, no chain-of-thought. A checkpoint-and-resume mechanism (every 50 questions, plus a crash-safety flush) tolerates rate-limit interruptions; a cap-aware job queue defers rate-limited work and resumes it after daily resets. Per-call token and wall-clock latency are recorded for cost/latency

TABLE IV
UPPER: SOLO ACCURACY OF EVERY MODEL IN THIS STUDY ($N = 80$).
LOWER: THE RESULTING PER-PAIR CAPABILITY GAP $|\text{CHEAP A} - \text{CHEAP B}|$ IN PP. BOLD = EXCEEDS THE ± 3 PP MATCHING BAR.

Model	ARC	MMLU	M-Pro	TQA
gpt-oss-120b (<i>frontier</i>)	95.0	82.5	71.2	78.8
gpt-oss-20b	96.2	82.5	65.0	71.2
gpt-oss-safeguard-20b	93.8	85.0	63.7	75.0
qwen3.6-27b	96.2	82.5	63.7	81.2
qwen3.8-27b	95.0	86.2	63.7	83.8
<i>Per-pair capability gap (pp)</i>				
Pair 3 (within)	2.4	2.5	1.3	3.8
Pair 7 (within)	1.2	3.7	0.0	2.6
Pair 10 (cross)	0.0	0.0	1.3	10.0
Pair 11 (cross)	1.2	3.7	1.3	12.6

OpenBookQA and CommonsenseQA were not part of the solo sweep; pair-level capability there is visible in the easy-tier Best-Cheap column of Table VI, where the four pairs span 88.6–93.8%.

TABLE V
C-HALL RATE ON AGREED QUESTIONS, BY BENCHMARK (MEAN OVER THE 4 COMPLETE PAIRS; RANGE ACROSS PAIRS IN PARENTHESES).

Benchmark	Mean C-Hall	Tier
OpenBookQA	3% (1.6–4.5)	easy
ARC-Challenge	3% (2.6–3.9)	easy
MMLU	10% (8.3–11.3)	medium
CommonsenseQA	10% (7.1–15.6)	easy
TruthfulQA	14% (9.5–19.8)	hard
MMLU-Pro	19% (13.3–25.9)	hard

analysis. All runs used free-tier APIs; total marginal API spend \$0.

G. Ablation design

All policy-ablation rows for a given (pair, benchmark) share one stored result set; only the decision rule differs. Always-Frontier accuracy is taken from the solo sweep (Table IV). PCR-3 is run separately ($N = 120$ /benchmark).

V. RESULTS AND DISCUSSION

A. Finding 1: C-Hall scales with difficulty, not task type

Prior work characterises coordinated hallucination as a *task-type* split — low on knowledge retrieval ($\approx 9\%$), high on mathematical reasoning ($\approx 32\%$) — and prescribes task-type routing accordingly. On the consistent 4-pair grid the effect is instead a smooth monotone climb with question difficulty, entirely *within* knowledge tasks (Table V, Fig. 1): the mean agreed-answer error rate rises from $\approx 3\%$ on ARC/OpenBookQA to $\approx 19\%$ on MMLU-Pro, and for the within-family Qwen pair (Pair 7) reaches **25.9% on MMLU-Pro** — the error rate previously attributed to mathematical reasoning, here on a 256-token multiple-choice task with $< 3\%$ unparsed responses. It is a property of question difficulty, not of the answer format or the task type.

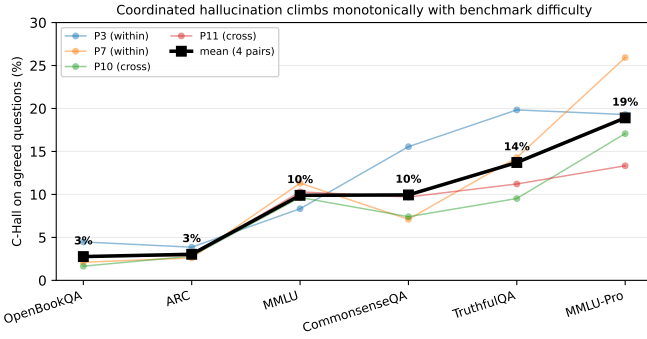


Fig. 1. Coordinated hallucination across the four pairs, all six benchmarks. The mean climbs monotonically with difficulty ($\approx 3\% \rightarrow 19\%$); the within-family Qwen pair (P7) reaches 25.9% on MMLU-Pro — an error rate previously reported only for mathematical reasoning, here on a plain multiple-choice task.

The prerequisite for safe PCR is difficulty-tier awareness, not task-type awareness. A trust threshold should be calibrated on question difficulty, which the escalation rate itself estimates (§fsec:earn). CommonsenseQA is the one non-monotone point ($\approx 10\%$, between MMLU and the easy pair): its C-Hall is driven by Pair 3, whose safety-tuned model produces 15.6% there.

B. Finding 2: Full policy ablation

Table VI scores eight policies by difficulty tier. **Easy:** on ARC every pair is within ≈ 1 pp of Always-Frontier (95.0%) and PCR-2’s only value is cost; CommonsenseQA is harder than its “easy” label suggests (C-Hall 7–16%), which is why the per-pair easy-tier means span 89–93%. **Medium:** PCR-2 adds +1 to +16 pp over Best-Cheap depending on cheap-model strength, and the two all-Groq cross-family pairs (10, 11) reach 86–88% — above Always-Frontier’s 82.5%. **Hard:** all four pairs land at 76.7–78.3% — *matching or beating* Always-Frontier (75.0%). But 12–20% of the answers PCR-2 trusts on the hard tier are wrong, so the accuracy is real and the safety is not; the within-family pairs (3, 7) carry the higher C-Hall (20%) versus the cross-family pairs (12–13%), consistent with Finding 5. **Oracle-2-cheap** (right if either cheap model is right) sits 3–5 pp above PCR-2 on several cells — a smarter-than-binary selector has real headroom.

C. Finding 3: Does a cheap pair beat the frontier?

Fig. 2 and Table VII give the measured trade-off over the 24 cells. **On easy and medium benchmarks PCR-2 matches or beats Always-Frontier accuracy at 22–72% lower cost** — the easy points sit up-left of the frontier $\times$ (same accuracy, one-third the cost) and all four MMLU points (84.0–87.6%) sit *above* the frontier $\times$ (82.5%). On MMLU-Pro all four PCR-2 points clear the frontier (71.2%) — three of them by 5 pp — and at lower cost, but 13–26% of what PCR-2 trusts there is wrong (Finding 2), so it wins on accuracy-per-dollar while losing on safety. On MMLU-Pro the cost saving shrinks to 22–46%.

TABLE VI

POLICY ABLATION BY TIER. AF = ALWAYS-FRONTIER (GPT-OSS-120B SOLO: ARC 95.0, MMLU 82.5, MMLU-PRO 71.2, TRUTHFULQA 78.8; HARD-TIER MEAN 75.0). PCR-2 ACC, BEST-CHEAP ACC, C-HALL ON AGREED SET, ESCALATION RATE — ALL %.

Tier	Pair	PCR-2	BestCh	AF	C-Hall	Esc.
easy	Pair 7	93.0	93.8	95.0	3.9	5.8
easy	Pair 10	91.2	93.6	95.0	4.0	11.1
easy	Pair 11	90.6	91.5	95.0	5.1	12.1
easy	Pair 3	89.4	88.6	95.0	8.0	8.5
medium	Pair 11	87.6	83.2	82.5	10.3	18.4
medium	Pair 10	86.0	82.8	82.5	9.6	21.2
medium	Pair 7	84.0	83.2	82.5	11.3	11.6
medium	Pair 3	84.8	69.2	82.5	8.3	42.4
hard	Pair 11	78.3	72.7	75.0	12.3	39.3
hard	Pair 10	78.0	73.7	75.0	13.3	37.7
hard	Pair 7	78.0	71.7	75.0	20.1	19.7
hard	Pair 3	76.7	72.7	75.0	19.6	21.7

Mean over each pair’s cells in that tier. Pair 3’s “easy” mean is pulled down by CommonsenseQA (15.6% C-Hall, 83% acc) — not as easy for the gpt-oss pair as ARC/OpenBookQA. Pair 3’s MMLU escalation is high (42%) because its safety-tuned model disagrees often.

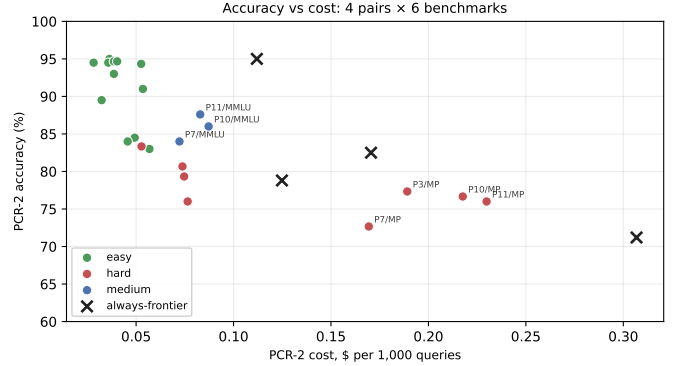


Fig. 2. PCR-2 accuracy vs. cost per 1,000 queries for the 4 complete pairs $\times$ 6 benchmarks, coloured by difficulty tier. $\times$ = Always-Frontier (gpt-oss-120B run solo) at its own cost and accuracy. Easy points are up-left of the frontier; the MMLU and MMLU-Pro points sit above it.

Latency is set by the slowest cheap model, not the policy: Pair 7 (both Qwen3 27B on Groq) runs PCR-2 at **0.4–0.6 s on every benchmark**, whereas Pairs 10/11 (which pair gpt-oss-20B with Qwen) run at **0.9–5.7 s** — gpt-oss-20B’s reasoning prelude, not the extra model or the provider, is the cost. For a latency-sensitive deployment the cheap pair should be two *non-reasoning* small models on the same fast provider.

D. Finding 4: PCR-3 does not help

A third cheap model (all-Groq trio, $N = 120/\text{benchmark}$) breaks nearly every tie (a 2-of-3 majority forms 84–100% of the time), collapsing escalation—but buys **no accuracy and no safety** (Table VIII). C-Hall is flat-to-worse on every benchmark: the items that stop escalating are added to the trusted set without being any cleaner. PCR-3 trades one frontier call for one extra cheap call per query—net-negative on easy/medium,

TABLE VII

MEASURED COST AND LATENCY FOR THE 4 COMPLETE PAIRS (PER-CALL INSTRUMENTATION; REPRESENTATIVE ROWS ORDERED BY COST CUT).
COST = LIST-PRICE TOKENS; RUNS WERE FREE-TIER. AF = ALWAYS-FRONTIER.

Pair / bench	Esc.	\$/1k PCR	\$/1k AF	cut	lat.
Pair 7 / OpenBookQA	5.0%	0.028	0.100	72%	0.49 s
Pair 7 / CommonsenseQA	8.5%	0.032	0.104	69%	0.55 s
Pair 7 / ARC	4.0%	0.036	0.108	66%	0.40 s
Pair 10 / OpenBookQA	9.0%	0.036	0.092	61%	4.31 s
Pair 7 / TruthfulQA	11.3%	0.053	0.130	59%	0.38 s
Pair 11 / MMLU	18.4%	0.083	0.172	52%	1.27 s
Pair 10 / MMLU	21.2%	0.087	0.170	49%	1.94 s
Pair 3 / MMLU-Pro	24.0%	0.189	0.349	46%	1.55 s
Pair 7 / MMLU-Pro	28.0%	0.169	0.287	41%	0.63 s
Pair 10 / TruthfulQA	30.0%	0.075	0.122	39%	1.03 s
Pair 11 / MMLU-Pro	50.0%	0.230	0.294	22%	3.06 s

TABLE VIII

PCR-3 (GPT-OSS-20B + QWEN3.6-27B + QWEN3.8-27B, MAJORITY VOTE) VS. PCR-2 (PAIR 7, THE SAME TWO QWEN MODELS). $N = 120$.

Benchmark	Policy	Esc.	C-Hall	Overall acc.
ARC	PCR-2	4.0%	2.6%	95.0%
	PCR-3	0%	5.0%	95.0%
TruthfulQA	PCR-2	11.3%	14.3%	83.3%
	PCR-3	1%	18.5%	81.7%
MMLU-Pro	PCR-2	28.0%	25.9%	72.7%
	PCR-3	16%	28.7%	73.3%

roughly break-even on hard. **Binary-agreement PCR-2 with a matched pair remains the right design.**

E. Finding 5: Model family — agreement premium on hard questions; a suggestive C-Hall premium there too

On the balanced 4-pair design (2 within, 2 cross, all 6 benchmarks, shared frontier and provider), bootstrap 95% CIs on the within-minus-cross gap (Table IX) split cleanly by difficulty. **Agreement.** On the easy benchmarks (ARC, OpenBookQA) both conditions agree ~ 90 – 96% and the gap is null. On the hard/adversarial benchmarks the within-family pairs agree far more: CommonsenseQA $+9.0$, TruthfulQA $+14.0$, MMLU-Pro $+21.7$ pp, every CI excluding 0. MMLU reverses (-7.2 pp, CI $[-12.2, -2.0]$) because Pair 3’s safety-tuned Cheap B refuses or hedges on many items, cutting that pair’s agreement to 58%. **Coordinated hallucination** follows the same shape, one step weaker: null on ARC, OpenBookQA, MMLU and CommonsenseQA (CIs span 0), but positive on the two hardest benchmarks — **TruthfulQA $+6.6$ pp, CI $[+0.3, +12.8]$** (excludes 0) and **MMLU-Pro $+7.2$ pp, CI $[-0.3, +14.9]$** (right at the boundary). The pair-level balanced gap is $+3.4$ pp. **Read conservatively: shared lineage buys a large agreement premium on hard questions, and plausibly a coordinated-hallucination premium there too — but with only two pairs per condition the C-Hall signal is a hypothesis for a powered replication, not a settled result.** On easy and medium questions neither effect is present; what

TABLE IX

WITHIN-FAMILY MINUS CROSS-FAMILY GAP ON THE BALANCED 4-PAIR DESIGN.

Benchmark	within	cross	gap	95% CI
<i>Agreement rate</i>				
ARC	95.4	94.0	+1.4	$[-1.8, +4.7]$
OpenBookQA	92.2	89.5	+2.7	$[-1.3, +6.8]$
MMLU	73.0	80.2	-7.2	$[-12.2, -2.0]^*$
CommonsenseQA	90.8	81.8	+9.0	$[+4.2, +13.5]^*$
TruthfulQA	84.7	70.7	+14.0	$[+7.7, +20.3]^*$
MMLU-Pro	74.0	52.3	+21.7	$[+14.0, +29.3]^*$
<i>Coordinated hallucination</i>				
ARC	3.4	2.8	+0.5	$[-2.1, +2.9]$
OpenBookQA	3.3	2.2	+1.0	$[-1.3, +3.4]$
MMLU	10.1	10.0	+0.2	$[-4.0, +4.3]$
CommonsenseQA	11.3	8.6	+2.7	$[-1.8, +7.3]$
TruthfulQA	16.9	10.4	+6.6	$[+0.3, +12.8]^*$
MMLU-Pro	22.5	15.3	+7.2	$[-0.3, +14.9]^*$

*CI excludes 0. Question-level bootstrap, 10,000 iterations, over the 4 complete pairs (within = Pairs 3, 7; cross = Pairs 10, 11). Two pairs per condition — treat the C-Hall column as suggestive.

dominates C-Hall everywhere is benchmark difficulty (Finding 1). Note also that on TruthfulQA the capability gap runs *against* this result — the cross-family pairs are the mismatched ones there (10–12.6 pp, Table IV), which should inflate *their* C-Hall, yet they show the lower rate.

F. Finding 6: Capability matching is what makes a pair usable

Binary agreement is only a quality signal when both cheap models are *comparably* capable. If one is materially weaker, its errors dominate the agreed set: on items the weak model cannot do, it frequently lands on the same plausible-but-wrong distractor the stronger model would only occasionally pick, and the pair “agrees” on a wrong answer. The resulting C-Hall measures a **skill gap, not shared blind spots** — it is not evidence about lineage, and a study that mixes matched and mismatched pairs will read the former as the latter.

We therefore adopted a ± 3 pp solo-accuracy matching bar as an *a priori* screening criterion (§IV-A); it is the single largest reason candidate pairs were dropped. Within the four pairs retained, the bar holds on ARC, MMLU and MMLU-Pro, and the one place it does not — TruthfulQA, where the cross-family pairs carry 10–12.6 pp gaps (Table IV) — biases against rather than toward our family conclusion (Finding 5).

The practical corollary for practitioners is blunt: **do not build a PCR pair from two models of visibly different strength**, and in particular not from two sizes of the same model series, which are mismatched by construction. Verify the gap on the target distribution before deploying, not after.

G. Per-pair, per-benchmark grid

All results use the **four pairs with a complete six-benchmark grid**: two within-family (Pair 3, gpt-oss + gpt-oss-safeguard; Pair 7, Qwen3.6 + Qwen3.8) and two cross-family (Pair 10, gpt-oss-20B + Qwen3.6; Pair 11, gpt-oss-20B

TABLE X

THE CONSISTENT 4×6 GRID, PCR-2. **TOP:** COORDINATED-HALLUCINATION RATE (%) ON AGREED QUESTIONS. **BOTTOM:** AGREEMENT RATE (%). “(w)” = WITHIN-FAMILY. FRONTIER = GPT-OSS-120B FOR ALL FOUR; ALL ON GROQ.

<i>Coordinated hallucination (%)</i>						
Pair	ARC	MMLU	M-Pro	TQA	OBQA	CSQA
3 (w)	3.9	8.3	19.3	19.8	4.5	15.6
7 (w)	2.6	11.3	25.9	14.3	2.1	7.1
10	2.8	9.6	17.1	9.5	1.6	7.4
11	2.9	10.3	13.3	11.2	2.8	9.7

<i>Agreement rate (%)</i>						
Pair	ARC	MMLU	M-Pro	TQA	OBQA	CSQA
3 (w)	95	58	76	81	90	90
7 (w)	96	88	72	89	95	92
10	95	79	55	70	91	81
11	93	82	50	71	88	82

+ Qwen3.8). All four hold the frontier constant (gpt-oss-120B) and run on one provider, so provider and frontier are controlled and only lineage varies (§IV-A).

Table X gives the 4×6 grid for the two quantities that carry the paper’s argument. Two patterns: (i) C-Hall tracks benchmark difficulty monotonically for every pair ($\sim 2\text{--}5\%$ ARC/OpenBookQA, $\sim 7\text{--}16\%$ MMLU/CommonsenseQA, $\sim 10\text{--}20\%$ TruthfulQA, $\sim 13\text{--}26\%$ MMLU-Pro); (ii) the within-family pairs sit at the high end of agreement on the hard/adversarial benchmarks but *not* on MMLU, where Pair 3’s safety-tuned Cheap B model refuses or hedges, cutting agreement to 58%.

H. When the second cheap model earns its call

Table XI decomposes PCR-2 against the strongest single cheap model (the policy a practitioner would fall back to) and shows the escalation rate that drives its cost. Three things stand out.

The value is concentrated on the hard tier. PCR-2 beats Best-Cheap by +10 to +11pp on MMLU-Pro for Pairs 7/10/11, and by +15.6pp on MMLU for Pair 3 (whose weak safety-tuned Cheap B drags Best-Cheap down while consensus still filters well). On ARC the gap is ± 1 pp.

On the “easy” knowledge benchmarks PCR-2 can lose. On OpenBookQA all four pairs come in 1–2.5pp *below* Best-Cheap, and the cross-family pairs are 2–4pp below on CommonsenseQA. When cheap models are already near ceiling, the $\sim 10\text{--}20\%$ of queries that escalate flip about as many correct answers to wrong as they rescue — 9 of the 24 cells are net negative, eight of them on the easy tier (the ninth is Pair 10 on TruthfulQA). **PCR-2 is a middle-of-the-difficulty-range tool:** skip it where a single cheap model already clears $\sim 92\%$.

Escalation rate is a free difficulty gauge. It rises monotonically with benchmark difficulty for every pair (4–7% ARC $\rightarrow$ 12–42% MMLU $\rightarrow$ 24–50% MMLU-Pro) without any labels or training — the router’s own disagreement signal is

TABLE XI

TOP: PCR-2 ACCURACY MINUS THE STRONGER CHEAP MODEL’S SOLO ACCURACY (PP) — POSITIVE = THE SECOND MODEL PLUS ESCALATION HELPS. **BOTTOM:** ESCALATION RATE (%). “(w)” = WITHIN-FAMILY.

<i>PCR-2 – Best-Cheap (pp)</i>						
Pair	ARC	MMLU	M-Pro	TQA	OBQA	CSQA
3 (w)	+0.7	+15.6	+3.3	+4.7	−1.0	+3.0
7 (w)	−0.5	+0.8	+10.7	+2.0	−2.0	0.0
10	−0.7	+3.2	+11.3	−2.7	−2.5	−4.0
11	+0.7	+4.4	+10.0	+1.3	−1.5	−2.0

<i>Escalation rate (%)</i>						
Pair	ARC	MMLU	M-Pro	TQA	OBQA	CSQA
3 (w)	5	42	24	19	10	10
7 (w)	4	12	28	11	5	8
10	5	21	45	30	9	19
11	7	18	50	29	12	18

an unsupervised estimate of query hardness that a deployment can log and act on.

I. Practitioner recommendations

The four complete pairs also settle the practitioner question directly. The recommended default, **Pair 7 (Qwen3.6-27B + Qwen3.8-27B $\rightarrow$ gpt-oss-120B, all Groq)**, holds 90–95% on easy, 84% on MMLU (vs. 82.5% for always-frontier) and 73–83% on the hard tier, at 0.4–0.6s and 41–72% below always-frontier cost. The cross-family alternative, **Pairs 10/11 (gpt-oss-20B + Qwen)**, matches it on accuracy — in fact edges ahead on MMLU (86–88%) and the hard tier (76–78%) because the two developers’ errors are less correlated — but runs 2–10 $\times$ slower and escalates more (Table XI). Choose Pair 7 when latency dominates, Pair 10/11 when the workload leans hard and independence matters.

VI. CONCLUSION

Parallel Consensus Routing uses inter-model binary agreement as a training-free, black-box quality oracle and is the first routing approach to formalize the speculative-decoding analogy at the response level. On a balanced four-pair design spanning a complete six-benchmark grid — every pair evaluated on every benchmark, frontier and provider held constant — with an eight-policy ablation, we establish: (1) coordinated hallucination scales monotonically with question difficulty ($\approx 3\%$ to $\approx 19\%$ mean, up to 26% for the within-family pair), not with task type; (2) a capability-matched cheap pair with PCR-2 matches always-frontier accuracy on easy/medium queries at 40–70% lower cost, but is unsafe on hard queries where 12–26% of trusted answers are wrong; (3) shared model lineage buys a large agreement premium on hard/adversarial benchmarks and, suggestively, a coordinated-hallucination premium there too (TruthfulQA +6.6pp) — underpowered at two pairs per condition, so a hypothesis for replication; capability matching, not lineage, is what makes a pair usable; (4) a third cheap model breaks ties but adds no

TABLE XII
RECOMMENDED CONFIGURATION BY GOAL (FREE-TIER, 2026).

Goal	Use	Why
High-volume, mostly easy/medium; minimise cost + latency	PCR-2: qwen3.6-27b + qwen3.8-27b → gpt-oss-120b (all Groq)	≈Best-Cheap / Always-Frontier accuracy; 41–72% cheaper; 0.4–0.6 s; 4–12% escalation outside MMLU-Pro (28% there); no truncation.
Hard-leaning workload; want error-independence	PCR-2: gpt-oss-20b + qwen3.8-27b → gpt-oss-120b (Pair 11, all Groq)	Different developers ⇒ less-correlated errors; +4 pp on MMLU, +10 pp on MMLU-Pro over Best-Cheap. Cost: 1–3 s, higher escalation.
Easy benchmarks where a cheap model already clears ~92%	Skip PCR; use the stronger cheap model alone	On Open-BookQA/CommonsenseQA PCR-2 is 1–4 pp below Best-Cheap (Table XI) — escalation flips as many right as it fixes.
Hard / adversarial queries	Do not use PCR consensus as a trust signal; route to frontier or gate by difficulty	Agreement is 10–26% wrong there; PCR-2 gains +3–11 pp on MMLU-Pro but ships confident errors and is only 22–46% cheaper.
Research: settle the family question	≥3 capability-matched pairs / condition, $N \approx 1000$ on MMLU-Pro	Only MMLU/MMLU-Pro pools are large enough; each pair must clear the ±3 pp bar.

accuracy or safety; (5) on free-tier APIs, two of {gpt-oss-20B, Qwen3.6-27B, Qwen3.8-27B} on a low-latency provider is the recommended default. The practical consequence is that a PCR deployment needs a extittdifficulty-calibrated trust threshold rather than a task-type rule: the router’s own escalation rate supplies that difficulty estimate for free.

Limitations

Pilot-scale N (150–300 per cell) gives per-benchmark C-Hall CIs of ±3–14 pp, and the family comparison rests on two pairs per condition — the hard-tier C-Hall premium (Finding 5) needs a powered replication before it can be claimed. Restricting to pairs with complete coverage buys internal consistency at the cost of breadth: all four pairs are served by one provider and share one frontier, so **provider and frontier are held constant but also unvaried** — we cannot say whether the findings transfer to a different frontier or a slower serving stack, and a capability-matched within-family pair on a second provider remains the key missing replicate. On TruthfulQA the two cross-family pairs exceed the ±3 pp matching bar (10–12.6 pp, Table IV); the bias runs against Finding 5 rather than for it, but the cell is not cleanly matched. Pair 3’s Cheap B is a safety-tuned model that refuses or hedges on ~40% of MMLU items, halving its agreement and inflating escalation without inflating accuracy — **safety-tuned models are poor PCR cheap models**, and it was retained only because it is the one available same-lineage partner for gpt-oss-

20B. Cosine similarity between raw responses is unavailable in this run (a library incompatibility), so we report binary letter agreement only. Always-Frontier is a single model (gpt-oss-120b). All runs used free-tier APIs; paid-tier cost profiles differ.

Future Work

A statistically powered family study (≈ 1000 MMLU-Pro questions $\times \geq 3$ matched pairs per condition, on at least two providers) would let the hard-tier C-Hall premium be *claimed* rather than merely suggested. The 3–5 pp Oracle-2-cheap gap motivates a lightweight learned selector beyond binary agreement. Extending the grid to a **numeric-answer benchmark** requires first disentangling answer extraction from the token budget: a truncated multi-step solution yields an intermediate value that a last-number parser scores as a disagreement, so apparent C-Hall on such benchmarks is budget-sensitive and must be calibrated before it can be compared against the multiple-choice grid. Online difficulty threshold learning [11] would remove the calibration corpus. Extending PCR to open-ended generation needs a semantic agreement criterion; applying it to retrieval-augmented generation [27], where grounded facts may still be co-hallucinated, is a high-value target.

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